Ray Pomponio

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Education

MS, Biostatistics May 2023
University of Colorado–Denver
Thesis: *Mistaken Identities Lead to Missed Opportunities: Testing for mean difference in partially matched data*
Marvin Porter Award for Outstanding Academic Achievement

Graduate Coursework, University of Pittsburgh December 2025
Introduction to Causal Inference (graduate-level biostatistics)

BS, Economics May 2018
University of Pennsylvania, Wharton School
Dean's List

Technical Skills

Programming: Python (Matplotlib, NumPy, Scikit-learn, SciPy, PyTorch), R (tidyverse, mice, data.table), SAS, Unix shell scripting

Machine Learning & AI: Deep learning (CNNs, brain age modeling), supervised/unsupervised ML, Monte Carlo simulation, cross-validation, model selection, image processing, computer vision

Statistics: Multivariate regression, Bayesian data analysis, causal inference, propensity scores, survival analysis, longitudinal data, missing data/multiple imputation, sample size/power calculation

Data Infrastructure: SQL, DBT, Git-based workflows, REDCap (including API automation), Microsoft Power BI, Power Automate, end-to-end clinical data pipeline design

AI-Assisted Workflows: Claude (Anthropic), GitHub Copilot; applied to scientific report drafting, slide narrative development, code generation and debugging, literature synthesis, and translating analytical outputs into non-technical communications

Operating Systems: Linux, macOS, high-performance computing environments (HPCs)

Research & Professional Experience

Data Scientist 2024 – Present
University of Pittsburgh / Children's Hospital of Pittsburgh

- Lead biostatistician on a **CDC-sponsored New Vaccine Surveillance Network** study tracking respiratory viruses across 8,000+ pediatric encounters; designed and executed the full clinical analytics pipeline from study design through publication in *mBio* (2026).
- Collaborated directly with **CDC program scientists** on study design, data governance, and public health interpretation, translating policy-relevant questions into epidemiological data collection instruments.
- Designed and managed end-to-end data infrastructure spanning multiple clinical systems (REDCap, EHR extracts, laboratory databases); built a **Power BI dashboard** ranked 834 of 3,571 reports across UPMC, with 80 views and 6 unique users in the past month, enabling greater security and faster time-to-analysis in routine reporting.

- Collaborated with clinical investigators to define research questions, then developed and validated **multivariate machine learning risk-factor models** for predicting severe respiratory illness in pediatric populations.
- Mentored junior faculty, fellows, and MPH student on statistical methods for infectious disease research, leading to 3 successful proposals, one thesis defense, and one K-award submission.
- Integrated LLM-assisted workflows (Claude, GitHub Copilot) into routine deliverables, including automated report drafting, slide deck narratives for CDC stakeholders, and code generation, accelerating turnaround on recurring analytic tasks.
- Automated recurring data query workflows, saving clinical coordinators an estimated **3–4 hours per week** in manual data retrieval and reporting.

Independent Statistical Consultant 2023 – 2024
Upwork.com

- Advised medical device clients on **FDA submission study design**; performed sample size and power calculations to justify endpoints and regulatory proposals.
- Conducted **customer segmentation analysis** for an e-commerce client using clustering methods, identifying high-value repeat-purchasing behaviors to inform retention strategy.

Graduate Research Assistant 2021 – 2023
University of Colorado–Denver, Department of Biostatistics & Informatics

- Co-authored **10+ peer-reviewed manuscripts** and conference abstracts across pulmonary medicine, critical care, and clinical epidemiology, spanning seven journals.
- Collaborated with multidisciplinary teams of clinicians, epidemiologists, and pulmonary/critical care specialists to translate clinical questions into testable statistical hypotheses and funding applications.
- Completed doctoral-level coursework in **machine learning and model selection**.

Data Analyst 2018 – 2020
University of Pennsylvania, School of Medicine
Davatzikos Lab (Center for Biomedical Image Computing and Analytics)

- Developed **neuroHarmonize**, an open-source Python package for harmonizing neuroimaging datasets across medical centers using ComBat-based statistical methods; now with 100+ GitHub stars and **500+ citations**.
- Built automated **machine learning pipelines for image processing and feature extraction** that reduced MRI preprocessing timelines from months to days for a multi-center consortium studying biomarkers of neurodegeneration.
- Applied **deep learning and computer vision methods** to large-scale MRI datasets (14,000+ individuals) to model brain age trajectories and disentangle neurodegeneration from typical aging patterns.
- Presented novel methodologies and technical results to interdisciplinary audiences including neurologists, radiologists, and biostatisticians at Penn, UCSF, and the iSTAGING consortium.

Grants & Funding

- New Vaccine Surveillance Network, Centers for Disease Control and Prevention. Role: Data Scientist/Statistician. PI: Dr. Marian Michaels. University of Pittsburgh, 2024–present.

Awards & Honors

- Certificate of Appreciation. Centers for Disease Control and Prevention, 2024–2026.
- Marvin Porter Award for Outstanding Academic Achievement. University of Colorado–Denver, Department of Biostatistics & Informatics, 2023.
- Dean's List. University of Pennsylvania, Wharton School, 2018.
- Hackathon Winner. Ryder Transportation customer analytics, Wharton Customer Analytics Initiative, 2018.

Selected Publications

Full list: ncbi.nlm.nih.gov/myncbi/raymond.pomponio.1/bibliography/public/

Neuroimaging & Harmonization

- **Pomponio R**, Erus G, Habes M, Doshi J, Srinivasan D, Mamourian E, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *NeuroImage.* 2020;208:116450.
- Bashyam VM, Erus G, Doshi J, . . . , **Pomponio R**, et al. MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14,468 individuals worldwide. *Brain.* 2020;143(7):2312–2324.
- Habes M, **Pomponio R**, Shou H, et al. The Brain Chart of Aging: machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer's & Dementia.* 2021;17(1):89–102.
- Nasrallah IM, Gaussoin SA, **Pomponio R**, et al. Association of intensive vs standard blood pressure control with MRI biomarkers of Alzheimer disease: secondary analysis of the SPRINT MIND randomized trial. *JAMA Neurology.* 2021;78(5):568–577.
- Chen AA, Srinivasan D, **Pomponio R**, et al. Harmonizing functional connectivity reduces scanner effects in community detection. *NeuroImage.* 2022;256:119198.

Clinical & Epidemiological Research

- **Pomponio R**, Peterson RA, Owusu M, et al. Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcoholism: Clinical and Experimental Research.* 2025;49(1):165–174.
- Pless LL, Damodaran L, **Pomponio R**, et al. Emergence and epidemiology of dominant variants of human metapneumovirus in the United States between 2016 and 2021. *mBio.* 2026;17(2):e0261925.
- Dunbar PJ, Peterson R, McGrath M, **Pomponio R**, et al. Analgesia and sedation use during noninvasive ventilation for acute respiratory failure. *Critical Care Medicine.* 2024;52(7):1043–1053.
- Smith JB, Peterson RA, **Pomponio R**, et al. Donor derived cell free DNA in lung transplant recipients rises in setting of allograft instability. *Frontiers in Transplantation.* 2024;3:1497374.

- Mayer M, Badesch DB, . . . , **Pomponio R**, et al. Impact of the COVID-19 pandemic on chronic disease management and patient reported outcomes in patients with pulmonary hypertension. *Pulmonary Circulation.* 2023;13(2):e12233.

Presentations & Teaching

- Presented thesis research to the Colorado School of Public Health, 2023.
- Lightning talk and poster presentation, Symposium on Data Science and Statistics, 2022.
- Seminar presentations on methods in biostatistics and data science, 3+ venues.
- Teaching Assistant, Longitudinal and Survival Analysis, University of Colorado–Denver, 2023. Led 10+ interactive office hours.
- Referee, *Imaging Neuroscience* and *NeuroImage* (5+ manuscripts reviewed).